Vue.js Programming Lab

All Programs — Source Code Only

Program 1: Student Task Planner

```
<template>
  <div class="task-planner">
    <h2>Student Task Planner</h2>
    <div class="input-group">
      <input
        v-model="newTask"
        @keyup.enter="addTask"
        placeholder="Enter a new study task..."
      />
      <button @click="addTask">Add Task</button>
    </div>
    <ul>
      <li v-for="(task, index) in tasks" :key="index">
        {{ task }}
        <button @click="removeTask(index)">Complete</button>
      </li>
    </ul>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const newTask = ref('');
const tasks = ref(['Read Chapter 4', 'Submit Math Assignment']);

const addTask = () => {
  if (newTask.value.trim() !== '') {
    tasks.value.push(newTask.value.trim());
    newTask.value = '';
  }
};

const removeTask = (index) => {
  tasks.value.splice(index, 1);
};
</script>
```

Program 2: Personal Expense Tracker

```
<template>
  <div class="expense-tracker">
    <h2>Personal Expense Tracker</h2>
    <div>
      <input v-model="itemName" placeholder="Expense name" />
      <input v-model.number="amount" type="number" placeholder="Amount" />
      <button @click="addExpense">Add Expense</button>
    </div>

    <h3>Total Expenditure: ${{ totalExpense }}</h3>

    <table border="1">
      <tr><th>Item</th><th>Cost</th></tr>
      <tr v-for="expense in expenses" :key="expense.id">
        <td>{{ expense.name }}</td>
        <td>${{ expense.cost }}</td>
      </tr>
    </table>
  </div>
</template>

<script setup>
import { ref, computed } from 'vue';

const itemName = ref('');
const amount = ref(0);
const expenses = ref([]);

const totalExpense = computed(() => {
  return expenses.value.reduce((sum, item) => sum + item.cost, 0);
});

const addExpense = () => {
  if (itemName.value && amount.value > 0) {
    expenses.value.push({
      id: Date.now(),
      name: itemName.value,
      cost: amount.value
    });
    itemName.value = '';
    amount.value = 0;
  }
};
</script>
```

Program 3: Library Book Manager

```
<template>
  <div>
    <h2>Library Book Manager</h2>
    <ul style="list-style-type: none; padding: 0;">
      <li
        v-for="(book, index) in books"
        :key="index"
        :style="{ color: book.isBorrowed ? 'red' : 'green' }"
      >
        <strong>{{ book.title }}</strong> by {{ book.author }} -
        <span>{{ book.isBorrowed ? 'Borrowed' : 'Available' }}</span>

        <button @click="toggleStatus(index)" style="margin-left: 10px;">
          {{ book.isBorrowed ? 'Return Book' : 'Borrow Book' }}
        </button>
      </li>
    </ul>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const books = ref([
  { title: 'Data Structures', author: 'Mark Allen', isBorrowed: false },
  { title: 'Vue.js Up & Running', author: 'Callum Macrae', isBorrowed: true },
  { title: 'Clean Code', author: 'Robert C. Martin', isBorrowed: false }
]);

const toggleStatus = (index) => {
  books.value[index].isBorrowed = !books.value[index].isBorrowed;
};
</script>
```

Program 4: Quiz Assessment Portal

```
<template>
  <div>
    <h2>Quiz Assessment Portal</h2>

    <div v-if="!quizCompleted">
      <h3>Question {{ currentQuestionIndex + 1 }} of {{ questions.length }}</h3>
      <p>{{ questions[currentQuestionIndex].text }}</p>
      <div
        v-for="(option, index) in questions[currentQuestionIndex].options"
        :key="index"
      >
        <button @click="submitAnswer(index)">{{ option }}</button>
      </div>
    </div>
    <div v-else>
      <h3>Quiz Completed!</h3>
      <p>Your Score: {{ score }} / {{ questions.length }}</p>
    </div>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const currentQuestionIndex = ref(0);
const score = ref(0);
const quizCompleted = ref(false);

const questions = [
  { text: 'What is the capital of France?', options: ['Rome', 'Paris', 'Berlin'], correct: 1 },
  { text: 'Vue.js is a framework for?', options: ['Backend', 'Database', 'Frontend'], correct: 2 }
];

const submitAnswer = (selectedIndex) => {
  if (selectedIndex === questions[currentQuestionIndex.value].correct) {
    score.value++;
  }
  if (currentQuestionIndex.value < questions.length - 1) {
    currentQuestionIndex.value++;
  } else {
    quizCompleted.value = true;
  }
};
</script>
```

Program 5: Event Registration System

```
<template>
  <div>
    <h2>Tech Conference Registration</h2>
    <form @submit.prevent="registerUser">
      <div>
        <label>Name: </label>
        <input v-model="formData.name" required />
      </div>
      <div>
        <label>Email: </label>
        <input v-model="formData.email" type="email" required />
      </div>
      <div>
        <label>Ticket Type: </label>
        <select v-model="formData.type">
          <option value="VIP">VIP</option>
          <option value="General">General</option>
        </select>
      </div>
      <button type="submit">Register Now</button>
    </form>
    <h3>Registered Attendees ({{ registrations.length }})</h3>
    <ul>
      <li v-for="(user, index) in registrations" :key="index">
        {{ user.name }} ({{ user.email }}) - {{ user.type }} Ticket
      </li>
    </ul>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const formData = ref({ name: '', email: '', type: 'General' });
const registrations = ref([]);

const registerUser = () => {
  registrations.value.push({ ...formData.value });
  formData.value.name = '';
  formData.value.email = '';
  formData.value.type = 'General';
};
</script>
```

Program 6: Habit Wellness Tracker

```
<template>
  <div>
    <h2>Weekly Habit Wellness Tracker</h2>
    <table border="1" style="border-collapse: collapse; text-align: center;">
      <tr>
        <th>Habit</th>
        <th v-for="day in weekDays" :key="day">{{ day }}</th>
      </tr>
      <tr v-for="habit in habits" :key="habit.id">
        <td style="text-align: left; padding: 5px;">{{ habit.name }}</td>
        <td v-for="(completed, index) in habit.days" :key="index">
          <input type="checkbox" v-model="habit.days[index]" />
        </td>
      </tr>
    </table>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const weekDays = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'];

const habits = ref([
  { id: 1, name: 'Drink 2L Water', days: [false, false, false, false, false, false, false] },
  { id: 2, name: 'Read 30 mins',   days: [false, false, false, false, false, false, false] },
  { id: 3, name: 'Workout',        days: [false, false, false, false, false, false, false] }
]);
</script>
```

Program 7: Mini E-Commerce Store

```
<template>
  <div style="display: flex; gap: 40px;">
    <div>
      <h2>Products</h2>
      <div v-for="product in products" :key="product.id" style="margin-bottom: 15px;">
        <h4 style="margin: 0;">{{ product.title }}</h4>
        <p style="margin: 5px 0;">${{ product.price }}</p>
        <button @click="addToCart(product)">Add to Cart</button>
      </div>
    </div>

    <div>
      <h2>Shopping Cart ({{ cart.length }})</h2>
      <ul>
        <li v-for="(item, index) in cart" :key="index">
          {{ item.title }} - ${{ item.price }}
        </li>
      </ul>
      <h3>Total: ${{ cartTotal }}</h3>
    </div>
  </div>
</template>

<script setup>
import { ref, computed } from 'vue';

const products = ref([
  { id: 1, title: 'Wireless Mouse',       price: 25 },
  { id: 2, title: 'Mechanical Keyboard',  price: 80 },
  { id: 3, title: 'USB-C Cable',          price: 10 }
]);

const cart = ref([]);

const addToCart = (product) => {
  cart.value.push(product);
};

const cartTotal = computed(() =>
  cart.value.reduce((sum, item) => sum + item.price, 0)
);
</script>
```

Program 8: Job Application Tracker

```
<template>
  <div>
    <h2>Job Application Tracker</h2>
    <label>Filter Status: </label>
    <select v-model="filterStatus">
      <option value="All">All</option>
      <option value="Applied">Applied</option>
      <option value="Interview">Interview</option>
      <option value="Rejected">Rejected</option>
    </select>
    <ul>
      <li v-for="job in filteredJobs" :key="job.id">
        <strong>{{ job.company }}</strong> - {{ job.role }}
        <span>({{ job.status }})</span>
      </li>
    </ul>
  </div>
</template>

<script setup>
import { ref, computed } from 'vue';

const filterStatus = ref('All');

const jobs = ref([
  { id: 1, company: 'Google',    role: 'Frontend Engineer',  status: 'Interview' },
  { id: 2, company: 'Amazon',    role: 'Full Stack Dev',     status: 'Applied'   },
  { id: 3, company: 'StartupX',  role: 'UI Developer',       status: 'Rejected'  },
  { id: 4, company: 'Microsoft', role: 'Software Engineer',  status: 'Interview' }
]);

const filteredJobs = computed(() => {
  if (filterStatus.value === 'All') return jobs.value;
  return jobs.value.filter(job => job.status === filterStatus.value);
});
</script>
```

Program 9: Study Notes Organizer

```
<template>
  <div>
    <h2>Study Notes Organizer</h2>
    <div style="margin-bottom: 20px;">
      <input v-model="noteTitle" placeholder="Topic Title" style="display:block; margin-bottom: 5px;"/>
      <textarea v-model="noteContent" placeholder="Enter study notes here..." rows="3"></textarea>
      <br><button @click="addNote">Save Note</button>
    </div>
    <div
      v-for="note in notes"
      :key="note.id"
      style="border: 1px solid #ccc; padding: 10px; margin-bottom: 10px;"
    >
      <h3 style="margin-top: 0;">{{ note.title }}</h3>
      <p>{{ note.content }}</p>
      <button @click="deleteNote(note.id)">Delete Note</button>
    </div>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const noteTitle = ref('');
const noteContent = ref('');
const notes = ref([]);

const addNote = () => {
  if (noteTitle.value && noteContent.value) {
    notes.value.push({
      id: Date.now(),
      title: noteTitle.value,
      content: noteContent.value
    });
    noteTitle.value = '';
    noteContent.value = '';
  }
};

const deleteNote = (id) => {
  notes.value = notes.value.filter(n => n.id !== id);
};
</script>
```

Program 10: Inventory Stock Monitor

```
<template>
  <div>
    <h2>Inventory Stock Monitor</h2>

    <ul style="list-style: none; padding: 0;">
      <li
        v-for="(item, index) in inventory"
        :key="index"
        :style="{
          padding: '10px',
          marginBottom: '5px',
          backgroundColor: item.quantity < 5 ? '#ffebee' : '#f1f8e9'
        }"
      >
        <strong>{{ item.itemName }}</strong> - Stock Level: {{ item.quantity }}

        <span v-if="item.quantity < 5" style="color: red; font-weight: bold; margin-left: 10px;">
          Low Stock!
        </span>

        <button @click="restock(index, 10)" style="float: right;">Restock +10</button>
      </li>
    </ul>
  </div>
</template>

<script setup>
import { ref } from 'vue';

const inventory = ref([
  { itemName: 'Laptop Chargers',   quantity: 15 },
  { itemName: 'HDMI Cables',       quantity: 3  },
  { itemName: 'Wireless Headsets', quantity: 8  },
  { itemName: 'Webcams',           quantity: 1  }
]);

const restock = (index, amount) => {
  inventory.value[index].quantity += amount;
};
</script>
```